

S.S.V.P.S's B. S. Deore Polytechnic, Dhule
Department of Computer Engineering
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SEMINAR AND PROJECT INITIATION COURSE

Sem : CO 5 K	**SPI – 315003**	**Date :**
Topic :	Sign Language to Text Translator	
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Contents

- Certificate
- Abstract

1. Introduction
2. Literature Review
3. Information to Seminar Topic
4. Advantages and Disadvantages
5. Conclusion
6. Project Initiation :

6.1 Description of problem statement
6.2 Scope and objectives
6.3 Stake holder
6.4 Seminar execution plan
6.5 Platform / Equipment / Resources identification

7. Bibliography and References

Abstract

- Sign language is the main way of communication for deaf and mute people, but most people do not understand it.
- This seminar studies a Sign Language to Text Translator that recognises hand gestures in real time using a normal webcam.
- Key hand landmarks are extracted using Google MediaPipe, and an LSTM network studies the sequence of frames to recognise moving gestures.
- The recognised gesture is shown as readable text and can also be spoken using Text-to-Speech (TTS).

1. Introduction

- Over 70 million people around the world rely on sign language.
- A big communication gap exists between the deaf/mute community and the general public.
- Human interpreters are costly and not always available.
- Computer Vision and Deep Learning can recognise hand gestures and convert them into text.
- Our solution: a low-cost, AI-powered virtual interpreter that needs only a standard webcam.

2. Literature Review

Sr. No	Approach Type	Technologies Used	Dataset	Key Objective	Limitations
1	HMM-based ASL recognition (Starner and Pentland, 1995)	Hidden Markov Models on video features	Small American Sign Language vocabulary	Real-time sign recognition from video	Small vocabulary; controlled setup
2	CNN-based gesture recognition (Pigou et al., 2014)	Convolutional Neural Network with Kinect	Kinect gesture data (Italian gestures)	Recognise sign gestures with CNN	Needs special depth-camera hardware
3	Hand tracking (Zhang et al., 2020 – MediaPipe Hands)	Palm detector and hand landmark model	Annotated hand images	Real-time 21-point 3D hand landmarks on device	Gives landmarks only, not meaning; occlusion errors

2. Literature Review

Sr. No	Approach Type	Technologies Used	Dataset	Key Objective	Limitations
4	Sequence learning (Hochreiter and Schmidhuber, 1997)	Long Short-Term Memory (LSTM)	Sequence learning tasks	Learn long-term dependencies in sequences	Needs enough sequence data and training time
5	Neural Sign Language Translation (Camgoz et al., 2018)	Encoder-decoder network with attention	RWTH-PHOENIX-Weather 2014T	Translate sign video into spoken-language text	Needs large annotated data; weather domain only
6	Sign Language Transformers (Camgoz et al., 2020)	Transformer, joint recognition and translation	RWTH-PHOENIX-Weather 2014T	End-to-end sign recognition and translation	High computing need; limited domain

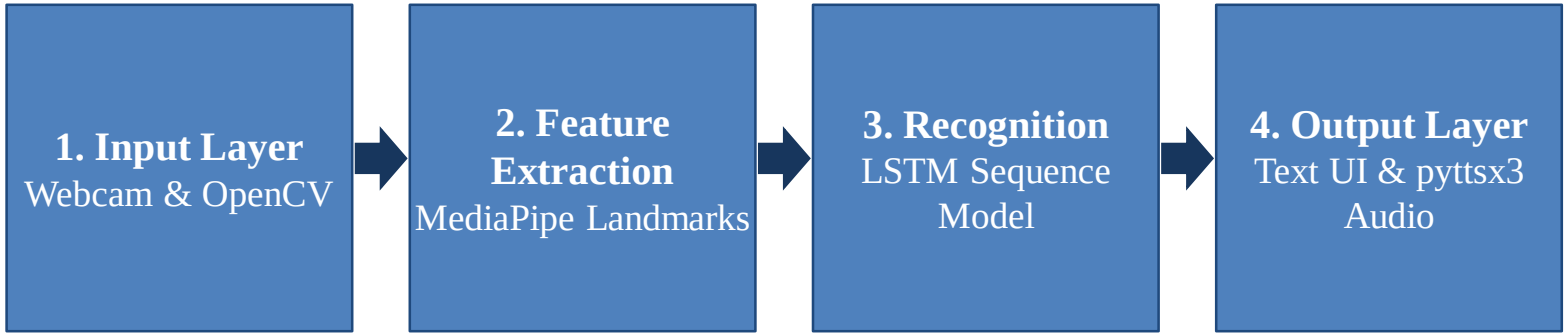
3.1 Overview of Sign Language Translation

- Sign language uses hand shapes, hand movement and facial expressions to communicate.
- A Sign Language to Text Translator captures signs through a camera and converts them into readable text (and speech).
- It acts as a low-cost virtual interpreter, so deaf/mute people can communicate with others without a human interpreter.
- Dynamic recognition: it understands moving words, not just static alphabet letters.

3.2 Technologies and Tools Used

Technology	Purpose
Python	Core programming language for scripting and model integration.
OpenCV	Real-time video capture, frame reading and image pre-processing.
MediaPipe	Extracts 21 3D hand landmarks (126 features for two hands) from each frame.
LSTM (TensorFlow)	Deep learning network that analyses a sequence of 30 frames over time.
pyttsx3	Offline Text-to-Speech library that converts recognised text into audio.

3.3 System Architecture



How it processes dynamic signs:

- The system keeps a sliding buffer of 30 consecutive frames; each frame has 126 coordinate values.
- This 30 × 126 matrix is passed to the LSTM, which predicts the moving gesture.

3.4 Working Principle

- **Step 1:** The webcam captures live video and OpenCV reads each frame.
- **Step 2:** MediaPipe detects the hands and extracts 21 3D landmarks per hand (126 features).
- **Step 3:** Landmark values of 30 consecutive frames are stored in a sliding buffer.
- **Step 4:** The LSTM model analyses the sequence and predicts the sign.
- **Step 5:** The recognised sign is displayed as text on the screen.
- **Step 6:** pyttsx3 converts the text into speech (offline).

3.5 Feature Extraction and Training Data

- **Why landmarks:** Using landmark coordinates instead of raw images makes the model light, fast and less affected by background.
- **Data collection:** Gesture sequences are recorded with the webcam for every sign to be recognised.
- **Training:** The 30-frame landmark sequences are used to train the LSTM (TensorFlow) model.
- **Testing:** Accuracy is checked on sample gestures that were not used in training.
- **Vocabulary:** The model recognises only the signs it was explicitly trained on.

4. Advantages / Disadvantages

Advantages	Disadvantages
Cost-Effective: needs only a standard webcam, no expensive sensor gloves.	Lighting Dependent: poor lighting reduces landmark accuracy.
Real-Time: MediaPipe and LSTM run smoothly on normal laptop CPUs.	Hand Occlusion: overlapping hands can confuse tracking.
Offline Audio: pyttsx3 gives speech without internet.	Limited Vocabulary: recognises only trained signs.
Dynamic Recognition: understands moving words, not just static letters.	No Facial Expressions: ignores non-manual grammar cues.

5. Conclusion

- The Sign Language to Text Translator helps reduce the communication gap between deaf/mute people and the general public.
- MediaPipe landmarks with an LSTM model allow real-time recognition of dynamic gestures using only a webcam.
- The system is low-cost and works offline, but depends on lighting and a limited trained vocabulary.
- In future, more signs, facial expressions and full sentence translation can be added.

6.1 Description of Problem Statement

- **Problem Statement:** A significant communication gap exists between the deaf/mute community and the general public. Human interpreters are costly and not always available.
- **Existing Systems:** Many solutions need expensive sensor gloves or special hardware.
- **Proposed Solution:** A low-cost, AI-powered virtual interpreter that recognises hand gestures through a standard webcam and converts them into text and speech.

6.2 Scope and Objectives

Scope

- Recognise dynamic hand gestures in real time using a webcam.
- Convert recognised signs into text and speech.
- Limited to the vocabulary of signs the model is trained on.
- Out of scope: facial expressions and full sign-language grammar.

Objectives

- Study existing sign language recognition methods.
- Extract hand landmarks using MediaPipe and OpenCV.
- Train an LSTM model on 30-frame gesture sequences.
- Display the result as text and audio output.
- Evaluate accuracy and note limitations for future work.

6.3 Stake Holder

Stakeholder	Role / Interest
Deaf / Mute Users	Use the system to communicate with others.
General Public	Understand signs without learning sign language.
Project Team	Designs, develops, tests and documents the system.
Schools, Hospitals and Service Centres	Can use it where interpreters are not available.
Project Guide and Evaluators	Guide the work and evaluate design, prototype and results.

6.4 Seminar Execution Plan

Week	Activity	Expected Output
Week 1	Selection of seminar topic and finalising the title	Topic approval
Week 2	Literature survey of sign language recognition work	Literature review notes
Week 3	Study of tools: OpenCV, MediaPipe, TensorFlow, pyttsx3	Tool list
Week 4	Designing architecture and working flow	Architecture diagram
Week 5	Collecting gesture data and building the prototype	Working prototype
Week 6	Training the LSTM and testing accuracy	Test results
Week 7	Preparing the seminar report and slides	Report and PPT draft
Week 8	Final presentation and submission	Final submission

6.5 Platform / Equipment / Resources

- **Programming:** Python.
- **Computer Vision:** OpenCV for video capture and pre-processing.
- **Hand Tracking:** Google MediaPipe (21 hand landmarks).
- **Deep Learning:** TensorFlow / Keras for the LSTM model.
- **Text-to-Speech:** pyttsx3 (offline).
- **Data Handling:** NumPy and Pandas.
- **Environment:** VS Code, Jupyter Notebook or Google Colab.
- **Hardware:** Laptop with webcam and at least 8 GB RAM.
- **Dataset:** Gesture sequences recorded by the team for the chosen signs.

7. Bibliography and References

[1] T. Starner and A. Pentland, “Real-time American Sign Language recognition from video using hidden Markov models,” 1995.

[2] L. Pigou et al., “Sign Language Recognition Using Convolutional Neural Networks,” ECCV Workshops, 2014.

[3] F. Zhang et al., “MediaPipe Hands: On-device Real-time Hand Tracking,” arXiv:2006.10214, 2020.

[4] S. Hochreiter and J. Schmidhuber, “Long Short-Term Memory,” Neural Computation, 1997.

[5] N. C. Camgoz et al., “Neural Sign Language Translation,” CVPR, 2018.

[6] N. C. Camgoz et al., “Sign Language Transformers: Joint End-to-end Sign Language Recognition and Translation,” CVPR, 2020.

[7] Documentation: OpenCV (opencv.org), TensorFlow (tensorflow.org), MediaPipe (mediapipe.dev), pyttsx3 (pypi.org/project/pyttsx3).

Thank you